M2.3: Transport Properties of Copper

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0.1 *Model Measurements*

0.1.1 *Electric Resistivity*

To investigat the electric resistivity a current ranging from $-100\,\mathrm{mA}$ to $100\,\mathrm{mA}$ in $20\,\mathrm{mA}$ steps is applied. The data points at ambient pressure are show in Figure 1 and in vaccuum in Figure 2. As there is the possiblity of some residue current, producing a voltage, a linear function is fittet. The y-axis shift is the the extra voltage produced by the resedure current. Both χ^2/ddof are close to 1, so the uncertainties are reasonable.

To calculate the electrical resistivity the extra current of $U_0^a = 9.29(22)\,\mathrm{nV}$ and $U_0^v = 7.09(25)\,\mathrm{nV}$ respectively are subtracted from the initial current. Thus the resulting current reads as formula

$$\rho = \frac{A(U_\mathrm{R}^a - U_0^a)}{I_\mathrm{R} l_\mathrm{R}} \tag{1}$$

with the distance between the two current probing contacts $l_\mathrm{R} = 48.8(20)\,\mathrm{mm}$, and the croess-sectional area $A = d \cdot b$ with $d = 0.111\,10(15)\,\mathrm{mm}$ and $b = 15.28(9)\,\mathrm{mm}$. The resistivity is shown in Figure 3 and Figure 4. The χ^2/ddof are notably low, which result from the fact, that the data points are nearly independent of the current I_E and thus lie all on the fit function. A fit is perhaps not appropriate here and thus the concept of a fit-quality breaks down.

Within the precision of the obtained data, it can be assumed that the electric resistivity is independent of the current. The data yields the electric resistivity $\rho^a = 18.46(26)\,\mathrm{n\Omega m}$ for ambient pressure and $\rho^v = 18.46(26)\,\mathrm{n\Omega m}$ in a vacuum. As both values are in a 6.46σ interval to the literature value $16.78\,\mathrm{n\Omega m}$[1] at 293 K, a systematic error is detected. This error can be caused by the contact resistance of the source contacts of the 4-point measurement.

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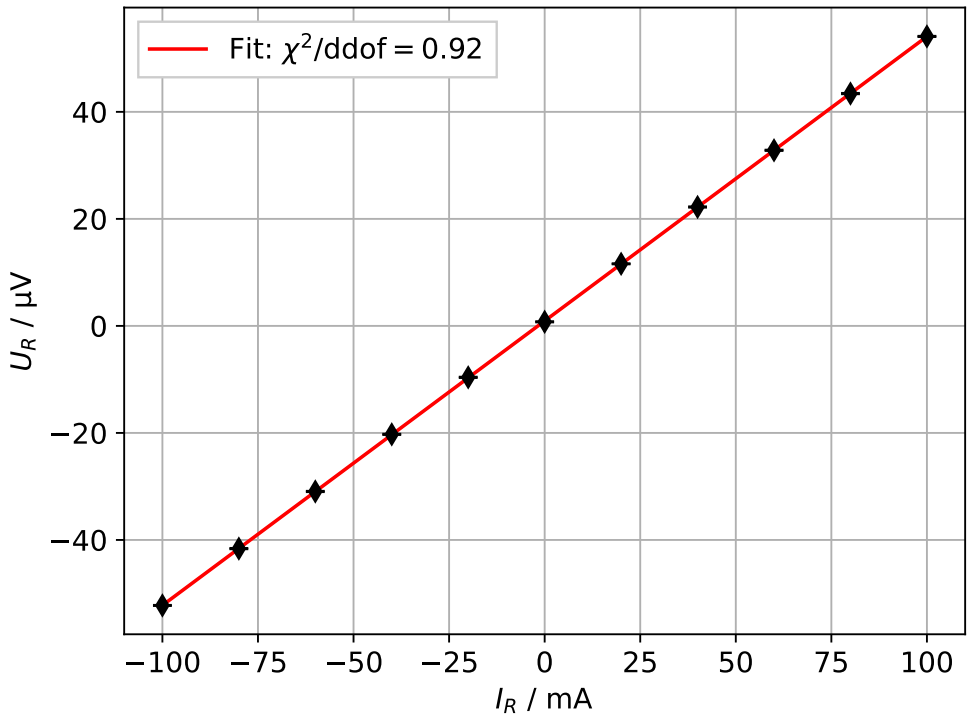


Figure 1: Electric Voltage at Ambient Pressure; Fit: $U_{\mathrm{R}}^{a}(I_{\mathrm{R}}) = 531.53(35)\,\mu\Omega \cdot I_R + 9.29(22)\,\mathrm{nV}$

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0.1.2 *Thermal Conductivity*

For the thermal conductivity a current ranging from 0 mA to 5 mA in 1 mA steps is used for the heater. The measured thermoelectric voltages are shown in Figure 5 and Figure 6.

Using the relation

$$\kappa = \frac{U_{\mathrm{PH}} I_{\mathrm{PH}} S(T,B) l_{\mathrm{LE}}}{U_{\mathrm{TE}} A} \tag{2}$$

with $U_{\mathrm{PH}} = R_{\mathrm{PH}} I_{\mathrm{PH}}$ Figure 7 and Figure 8 are obtained.

Here it can be seen that the thermal conductivity is dependent on the heating current at ambient pressure. In the vacuum the dependency is largely lost. This bevhavioure stems from the air around the sample. It can transport the heat through convection, which is a process with a dependent on the temperature, i.e. the heating current. As we approach the vacuum this contribution vanishes, as reflected in the data. Inspecting the rough ranges of thermal conductivity of both figures, one sees that the values are in general larger at similar currents in the air measurement then

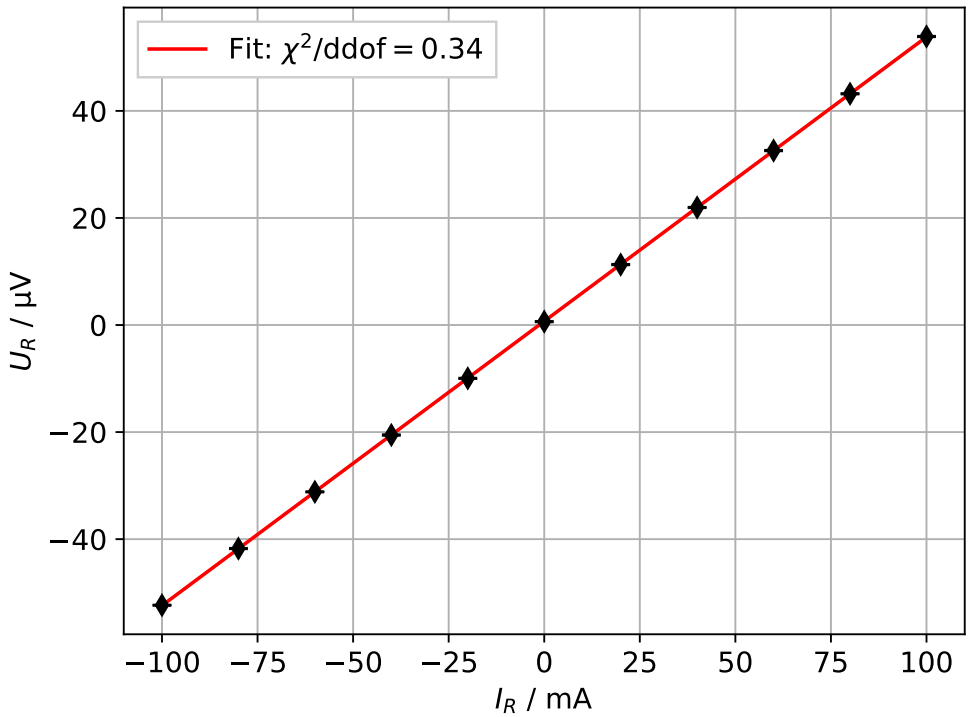


Figure 2: Elecgric Voltage in a Vacuum; Fit: $U_{\mathrm{R}}^{v}(I_{\mathrm{R}}) = 531.11(38)\,\mu\Omega{\cdot}I_{\mathrm{R}} + 7.09(25)\,\mathrm{nV}$

in the vacuum measurement. This is also due to the air which helps to transport the heat, thus making the thermal conductivity larger.

Unexpected is the observation, both thermal conductivities lie above the literature value $\kappa_{\mathrm{lit}} = 401\,\frac{\mathrm{W}}{\mathrm{mK}}$. Expected would be that the data would be below the literature value, since copper is in general very heat conductive and any building elements in the apparatus would rather be more insulating then have a higher thermal conductance. Therefore an unknown systematical error may apply.

Assuming the vacuum measurement to be independent of the heating current I_{PH}, the obtained value is with $604.0(130)\,\frac{\mathrm{W}}{\mathrm{m{\cdot}K}}$ about 15.6σ away from the literature value. This Also hints a systematical error. If the measured conductance would have been lower then the literature value, this discrepancy could have been explained by the contacts, which might be not as conducting as the copper. But as we have higher conductances, the authors can not think of a plausible explanation.

Lastly the temperature gradient is calculated via

$$\nabla T = \frac{\Delta T}{l_{\mathrm{TE}}} = \frac{U_{\mathrm{TE}}}{S(T,B) l_{\mathrm{TE}}} \tag{3}$$

with the thermoelectric power given by $S = 59\,\frac{\mathrm{mV}}{\mathrm{K}}$. The values are shown in Figure 9 and Figure 10

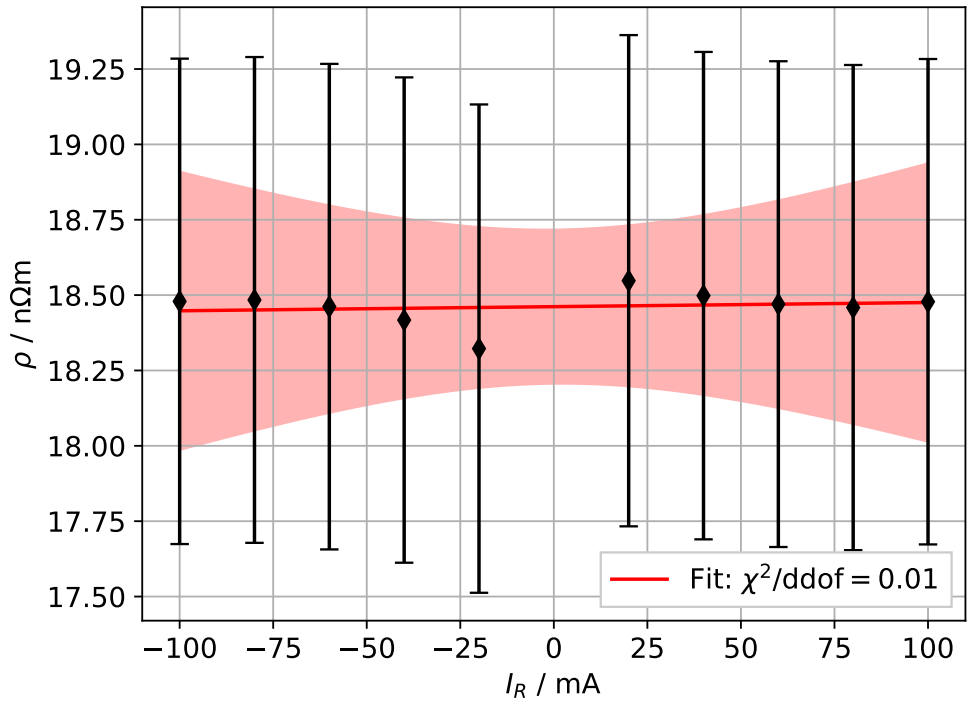


Figure 3: Electrical Resistivity at Ambient Pressure; Fit: $\rho^a(I) = 0.1(38)\,\frac{\mathrm{n\Omega m}}{\mathrm{A}} \cdot I + 1846(26)\,\mathrm{n\Omega m}$

As one can see, a cubed dependency on the I_{PH} is measured. Thus squaring the data beforehand could be a reasonable plot. But to as the square root of the temperature gradient is not commonly used, this report will keep the simple non-square root representation of the data.

This cubed behaviour is expected as the heating power is dependent on the heater current squared.

Just as discussed for the electric resistivity and as for the thermal conductivity, both measured values for ρ and κ are rather far from the literature value, but since their magnitudes are very comparable, all measured values are reasonable.

0.2 *Resistivity*

For the temperatures 10 K, 20 K, 30 K and 40 K and for the magnetic field strengths of −6 T to 6 T the data is recorded with a prepared dip-stick and are controlled by a computer. These measurements run overnight and are provided to us. The resistivity is calculated as before, but now with the values $d = 0.1111(15)$ mm, $b = 2.16(5)$ mm, $l_{\mathrm{TC}} = 16.2(5)$ mm, $l_{\mathrm{R}} = 12.2(5)$ mm, and $R_{\mathrm{PH}} = 1\,\mathrm{k\Omega}$. The resistivity is shown in Figures 11 to 14. The voltages are recorded for different current polarities. They are indicated with (+)$\hat{=}$1 A and (−)$\hat{=}$ − 1 A. As can be seen, the calculated resistivity

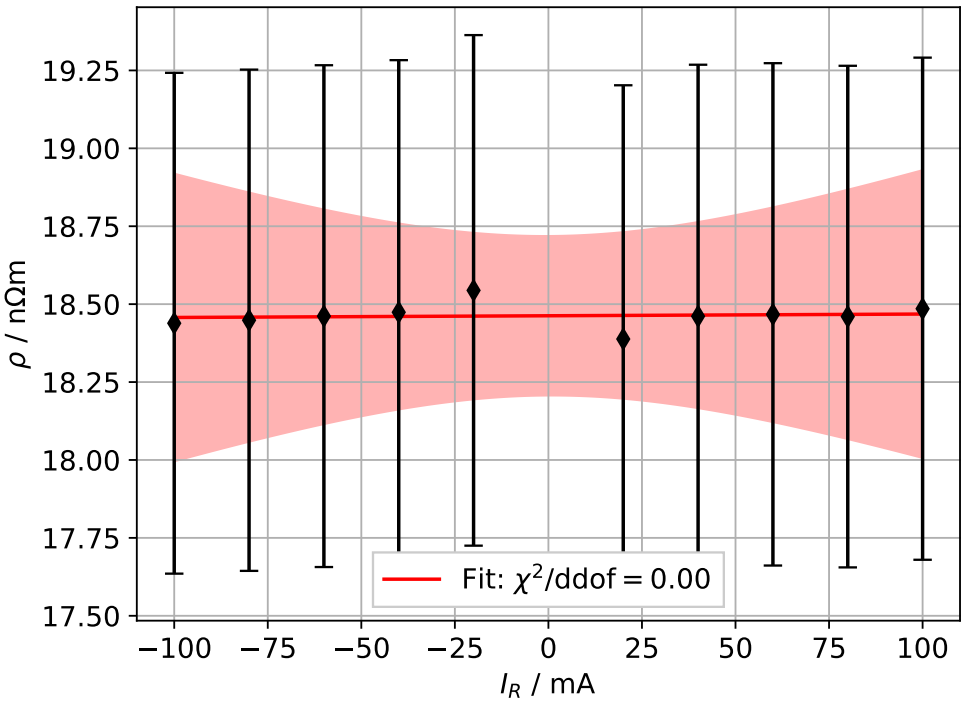


Figure 4: Electrical Resistivity in a Vacuum; Fit: $\rho^a(I) = 0.1(38)\,\frac{\mathrm{n\Omega m}}{\mathrm{A}} \cdot I + 18.46(26)\,\mathrm{n\Omega m}$

is different for B and $-B$ and for $(+)$ and $(-)$. In the case of the magnetic field, the difference comes from the fact, that the probing contacts are not perfectly centered and the magnetic field induces the hall voltage, which thus contaminates the measured voltage and produces an asymmetry in the measurement. In the case of the different polarities, the difference originates from a residue current, which induces a permanent voltage. In the following analysis, if not shown both, magnetic field and polarity, the average has been takes over the respectively not shown quantity. This cancels the asymmetry:

$$\begin{aligned} V_{\text{corrected}}(B) &= \frac{V(+B) + V(-B)}{2} \\ &= \frac{V(+B_0) + V(B_{\text{res}}) + V(-B) - V(B_{\text{res}})}{2} \\ &= \frac{V(+B_0) + V(-B_0)}{2} \end{aligned}$$

For high temperatures the and at zero-field, the resistivity is linear as shown in Figure 15. The low quality of $\chi^2/\text{ddof} = 0.01$ is due to the large uncertainties. They mainly come from the uncertainty of the geometry, like cross-sectional are and length.

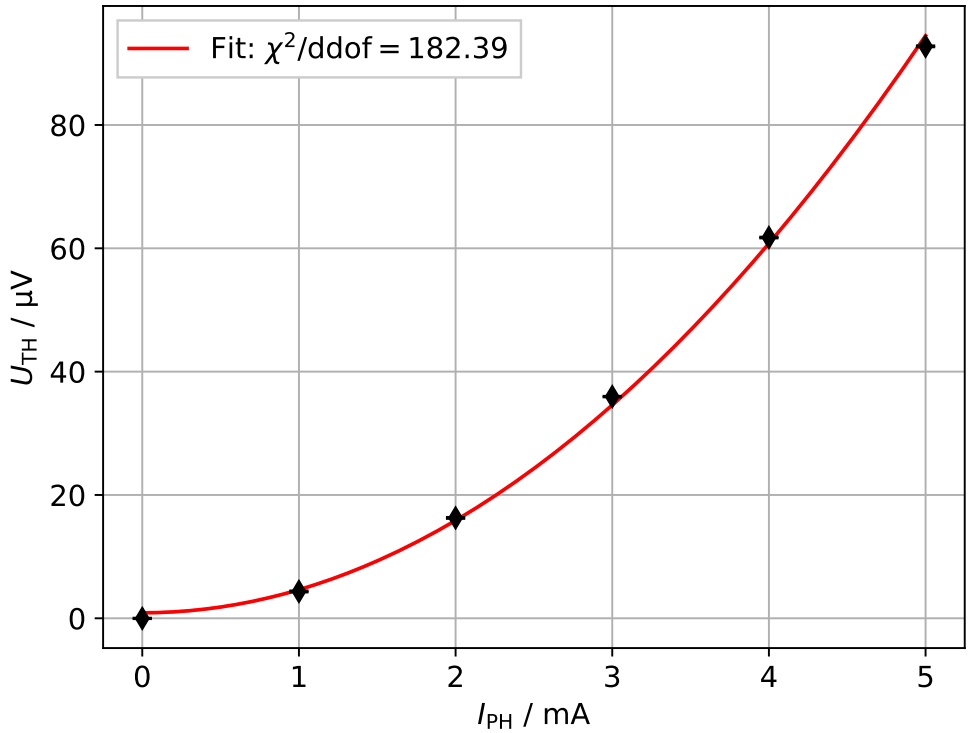


Figure 5: Thermalelectric voltage at Ambient Pressure; Fit: $U_{\mathrm{TH}}(I_{\mathrm{R}}) = 3.7434(48)\ \frac{\mathrm{V}}{\mathrm{I}^2} I_{\mathrm{PH}}^2 + 890(56)\,\mathrm{nV}$

I To determine the residual resistivity ration (RRR), the resistivity at 300 K is devided by the resistivity at 0 K in zero-field. Since the lowest resistivity is for 10 K, a fit is done with a displaced power law $\rho(T) = 66.59(93)\,\mathrm{f\Omega m} + 2.810(79) \times 10^{-18}\ \frac{\Omega\mathrm{m}}{\mathrm{K}^5} \cdot T^5$. As a result a RRR of 293(14) is obtained. As the application is at cryogenic temperatures, this copper should be comparable for OFHC copper, which usually used in this case. A RRR for normal OFHC Copper ranges from 50 to 75 [2]. Thus this copper is unusual pure for probably being OFHC copper.

The residual resistivity $\rho(T = 0\,\mathrm{K})$ is now subtracted and plotted in Figure 17. It can be seen that the power law of T^5 can model the behaviour very well for low temperatures. The quality of $\chi^2/\mathrm{ddof} = 1.54$ is close to 1, so the model works and the uncertainties are appropriate.

Now instead of plotting for different magnetic fields, the data is plotted for different temperatures. The result is seen in Figures 18 to 21. It can been that the resistivity has no preference for the direction of the magnetic field, except for the hall voltage contamination, and is thus symmetric. Rather strongly prominent is the difference for different current directions, showing that a non-neglectable residue current is present.

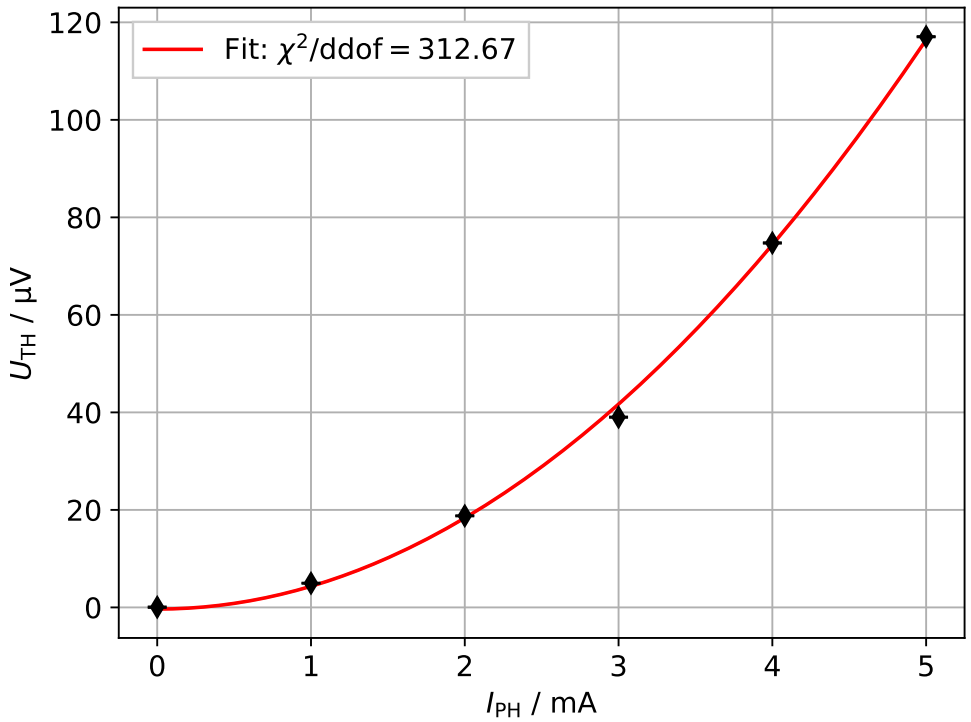


Figure 6: Thermalelectric voltage in a Vacuum; Fit: $U_{\mathrm{TH}}(I_{\mathrm{R}}) = 4.6703(38)\,\frac{\mathrm{V}}{\mathrm{I}^2} I_{\mathrm{PH}}^2 + -343(47)\,\mathrm{nV}$

Lastly the data can be presented in the Kohler plot. Here the relation

$$\frac{\Delta\rho}{\rho_0} = F\left(\frac{B}{\rho}\right) \tag{4}$$

with $\Delta\rho = \rho - \rho_0$ and $\rho_0 = \rho(T = 0\,\mathrm{K})$ is used. ρ_0 is also sometimes defined with $\rho(T = 4.2\,\mathrm{K})$ but since we need to extrapolate in either way, since we only have resistivities for as low as 10 K, we use the resistivity at 10 K. The Kohler rule says, that relation is independent of the temperature. As seen in Figure 22, this relation holds. The function F is material dependent and, taking an weighted average over all temperatures, can be determined to be $46.7(16)\,\frac{\Omega\mathrm{m}}{\mathrm{T}}$. All fit-qualities are far below 1, so the uncertainties are overestimated.

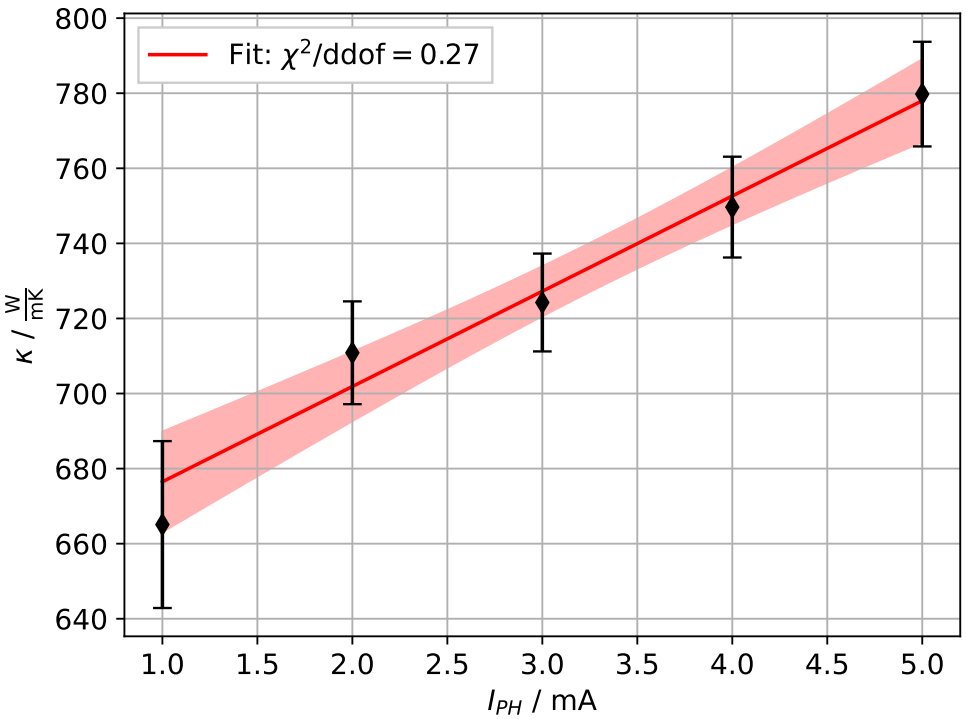


Figure 7: Thermal conductivity at ambient pressure: $\kappa(I_{\mathrm{PH}}) = 25\,400.0(51000)\ \frac{\mathrm{kW}}{\mathrm{m\cdot K\cdot A}} + 651.0(180)\ \frac{\mathrm{W}}{\mathrm{m\cdot K}} I_{\mathrm{PH}}$

Thermal conductivity at ambient pressure.			
mA	µV	mA	µV
0.0	−0.020(76)	3.0	35.951(86)
1.0	4.35(12)	4.0	61.75(10)
2.0	16.28(12)	5.0	92.75(12)
Thermal conductivity in vacuum.			
mA	µV	mA	µV
0.0	0.04(11)	3.0	39.017(83)
1.0	4.961(70)	4.0	74.750(95)
2.0	18.784(71)	5.0	117.052(82)

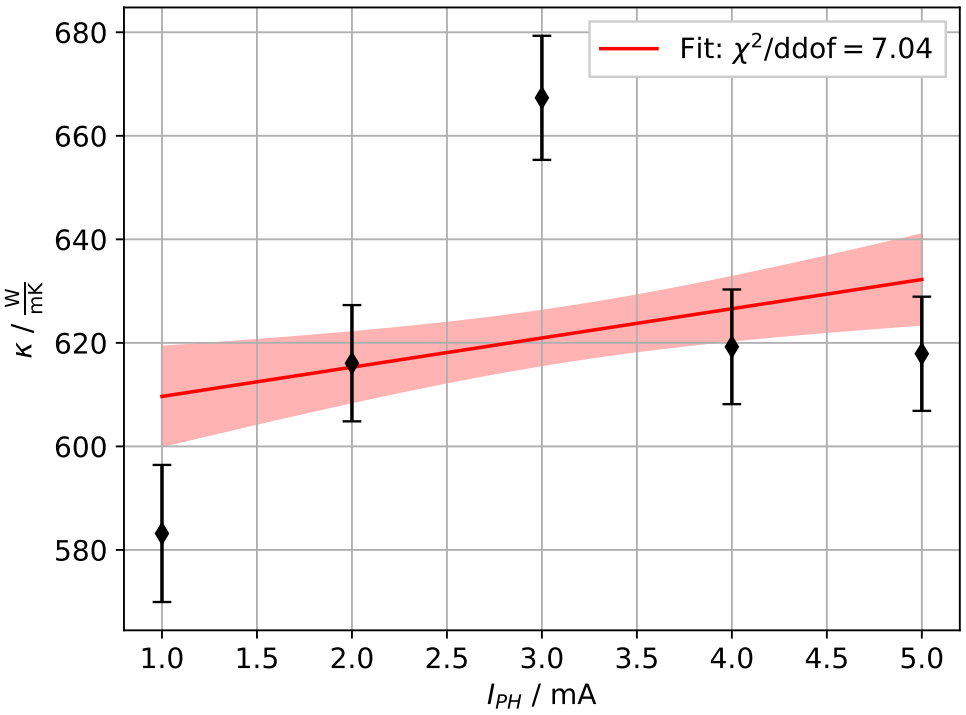


Figure 8: Thermal conductivity in a vacuum: $\kappa(I_{\mathrm{PH}}) = 5.6(38)\,\frac{\mathrm{kW}}{\mathrm{m\cdot K\cdot A}} + 604.0(130)\,\frac{\mathrm{W}}{\mathrm{m\cdot K}} I_{\mathrm{PH}}$

Electric resistivity measurement at ambient pressure.			
mA	µV	mA	µV
−100.0	−52.240(63)	20.0	11.602(61)
−80.0	−41.617(80)	40.0	22.218(77)
−60.0	−30.942(78)	60.0	32.814(79)
−40.0	−20.267(78)	80.0	43.417(82)
−20.0	−9.615(80)	100.0	54.094(70)
0.0	0.765(71)		
Electric resistivity measurement in vacuum.			
mA	µV	mA	µV
−100.0	−52.344(74)	20.0	11.290(85)
−80.0	−41.755(79)	40.0	21.955(81)
−60.0	−31.162(71)	60.0	32.589(89)
−40.0	−20.553(96)	80.0	43.200(82)
−20.0	−9.963(79)	100.0	53.895(84)
0.0	0.633(81)		

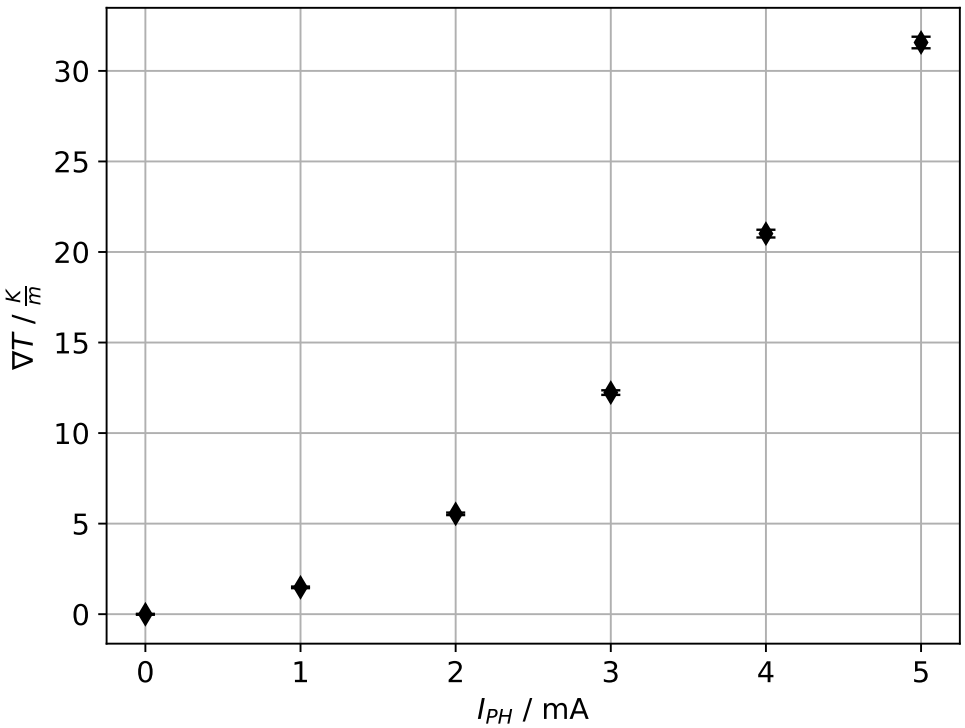


Figure 9: Temperature Gradient at Ambient Pressure.

References

[1] Rumble, J.R. (ed.): CRC Handbook of Chemistry and Physics, 105th edn. CRC Press, Boca Raton, FL (2024)

[2] Valois, J., Nellis, G., Pfotenhauer, J.: Characterization of the thermal properties of ofhc copper at cryogenic temperature. IOP Conference Series: Materials Science and Engineering **1301**, 012167 (2024) https://doi.org/10.1088/1757-899X/1301/1/012167

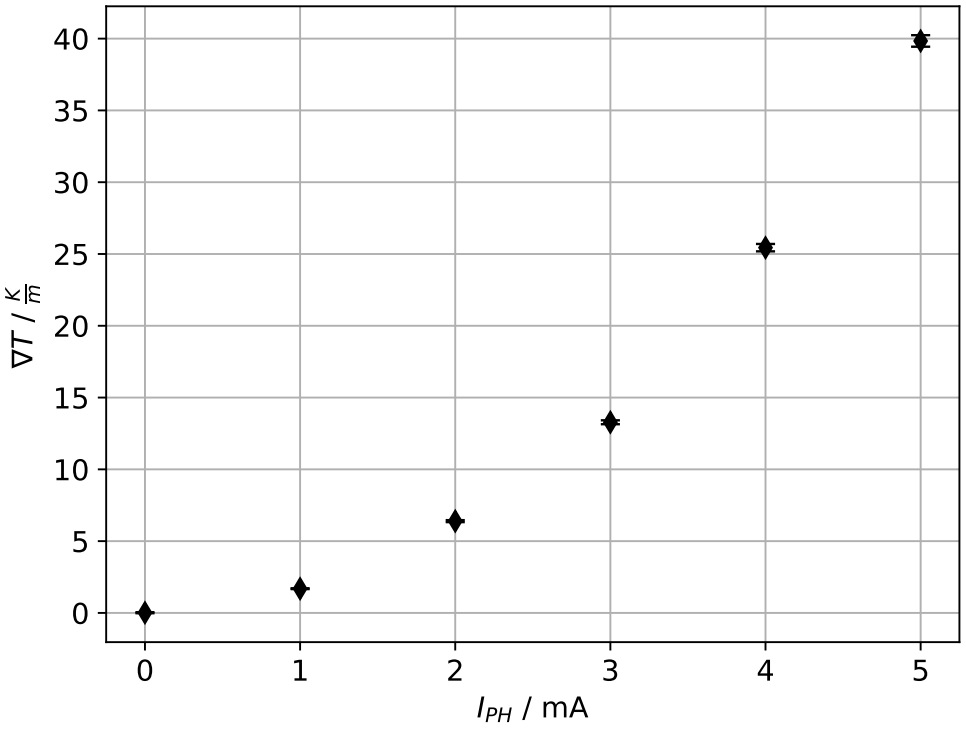


Figure 10: Temperature Gradient in a Vacuum

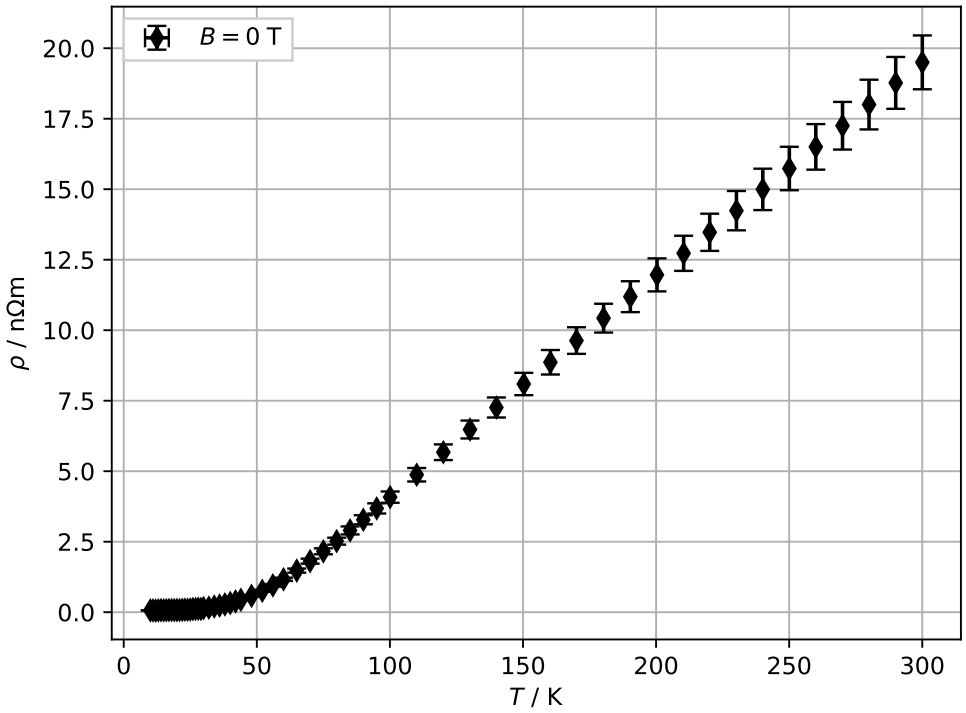


Figure 11: Electric resistivity for different temperatures at zero-field.

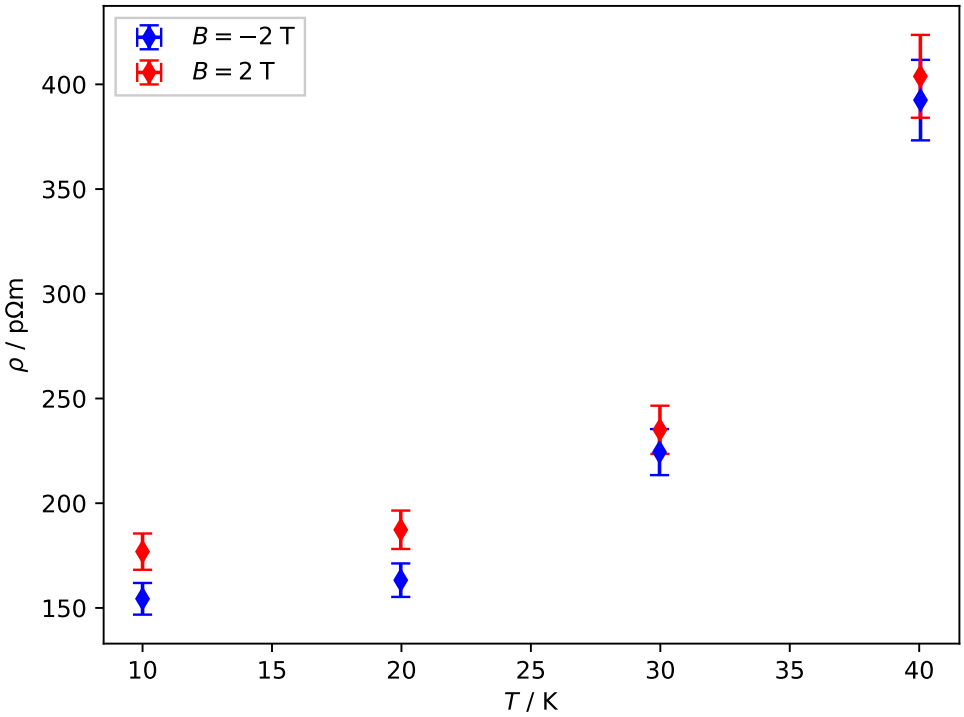


Figure 12: Electric resistivity for different temperatures in a magnetic field at 2 T and −2 T.

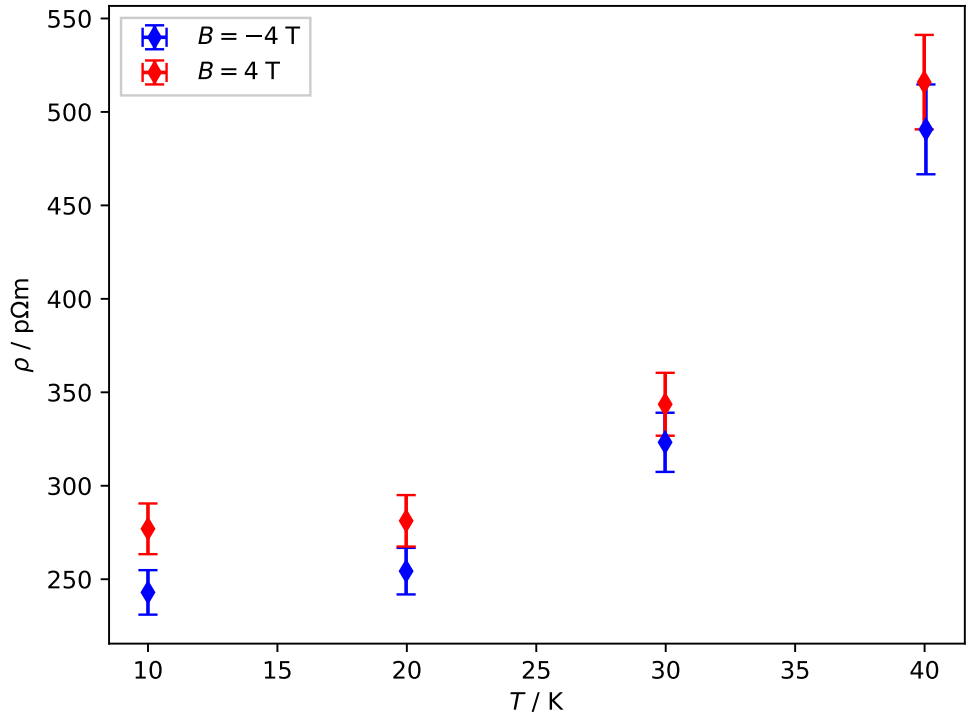


Figure 13: Electric resistivity for different temperatures in a magnetic field at 4 T and −4 T.

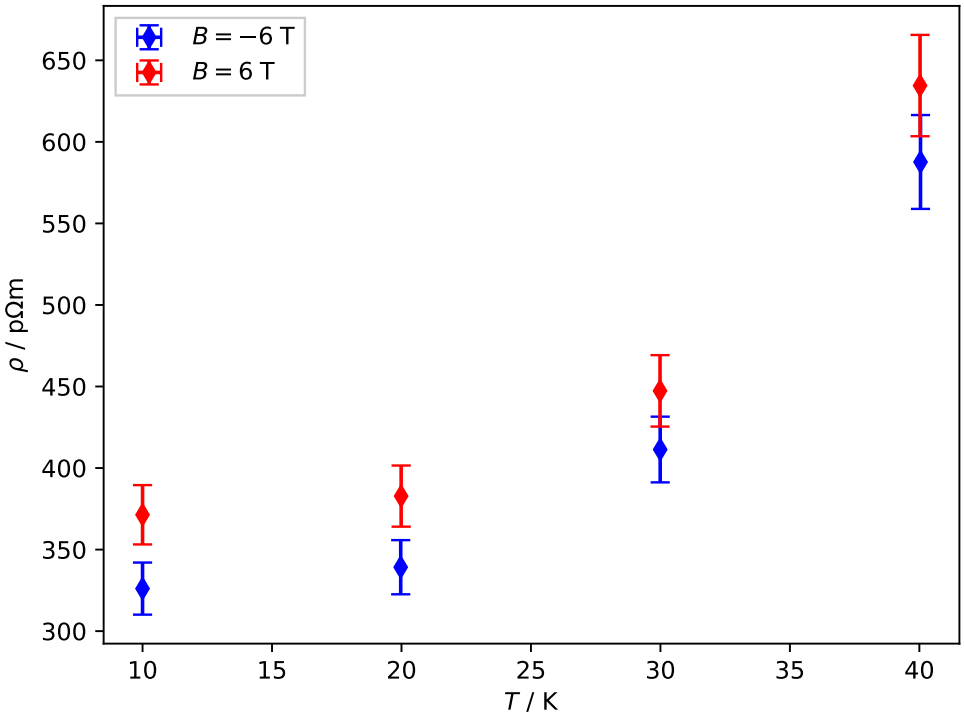


Figure 14: Electric resistivity for different temperatures in a magnetic field at 6 T and −6 T.

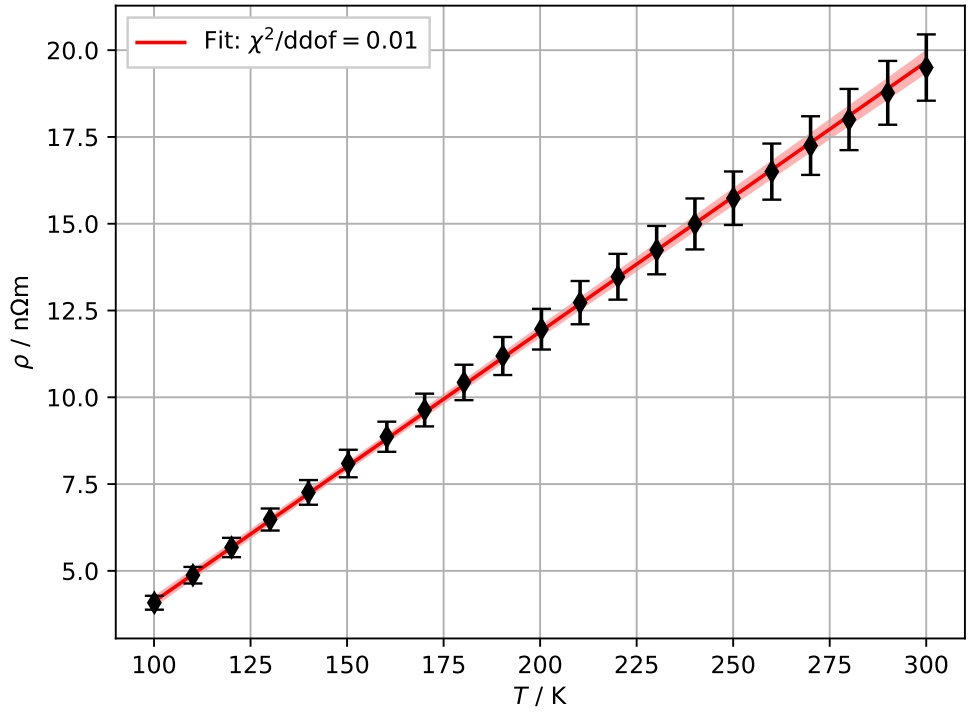


Figure 15: Resistivity for hight temperatures at zero-field.Fit: $\rho(T) = 77.8(18)\,\mathrm{f\Omega m} + -3.66(28)\,\frac{\mathrm{n\Omega m}}{\mathrm{T}} \cdot T$

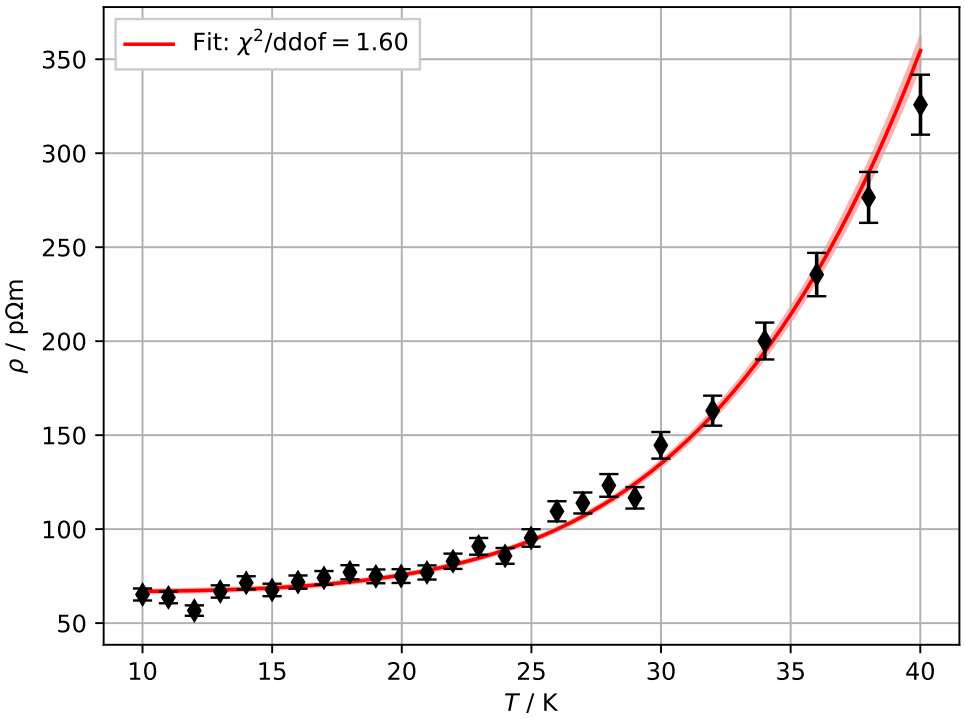


Figure 16: Resistivity for low temperatures at zero-field. Fit: $\rho(T) = 66.59(93)\,\mathrm{f\Omega m} + 2.810(79) \times 10^{-18}\,\frac{\Omega\mathrm{m}}{\mathrm{K}^5} \cdot T^5$.

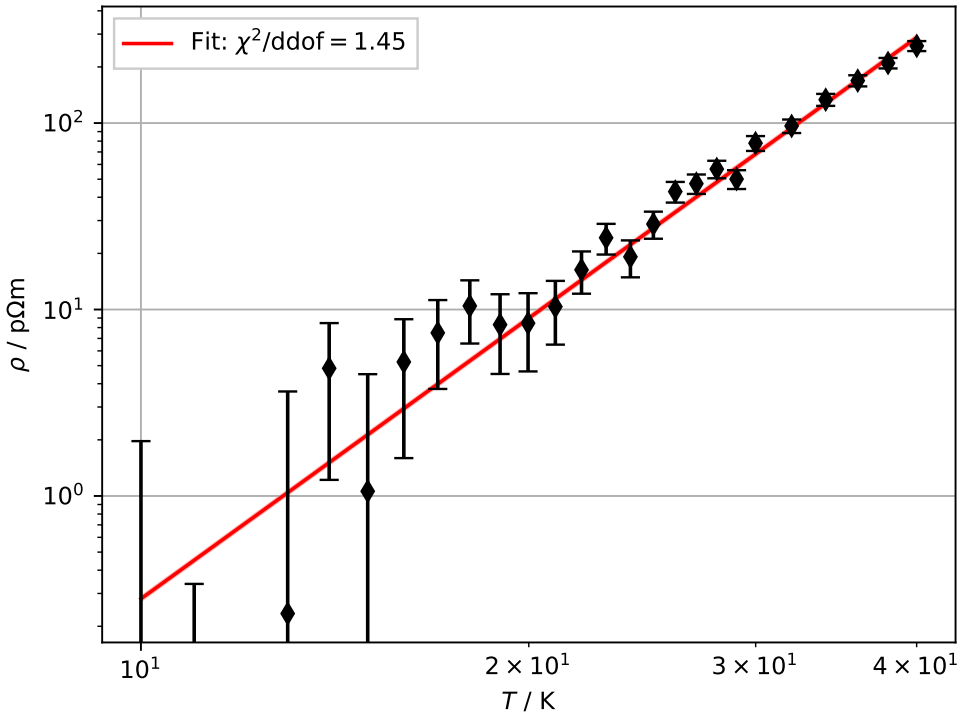


Figure 17: at zero-field

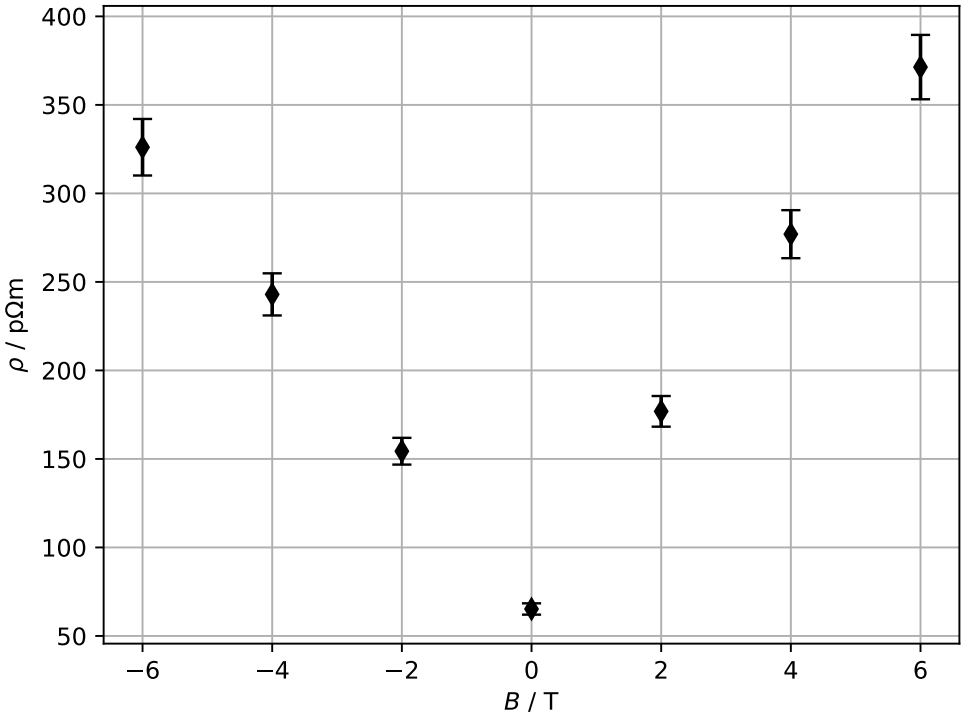


Figure 18: Electric resistivity for different magnetic fields at 10 K

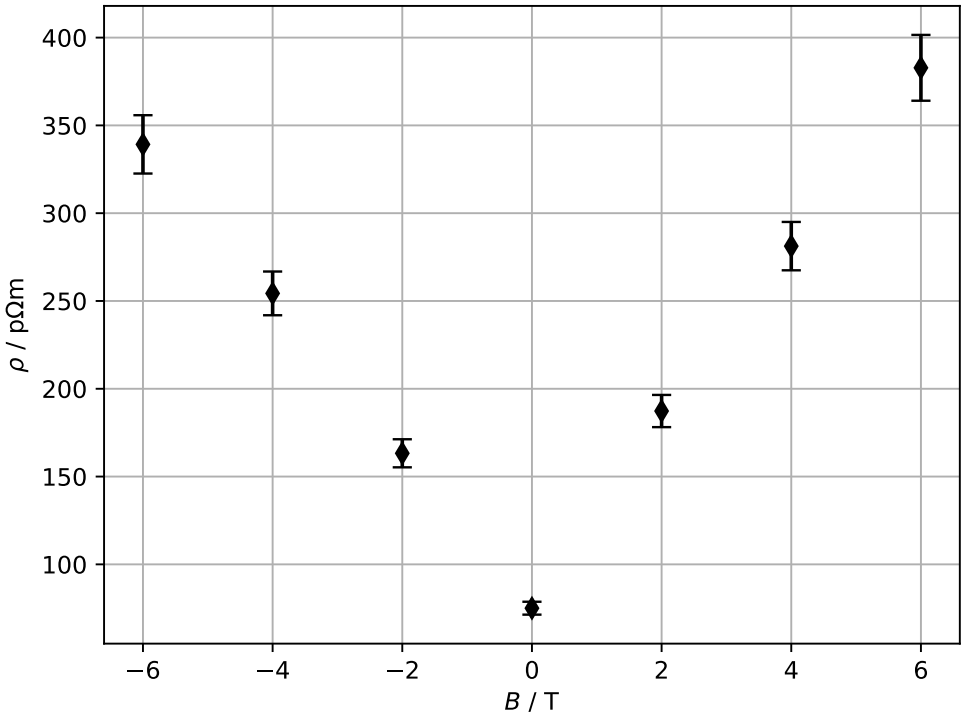


Figure 19: Electric resistivity for different magnetic fields at 20 K.

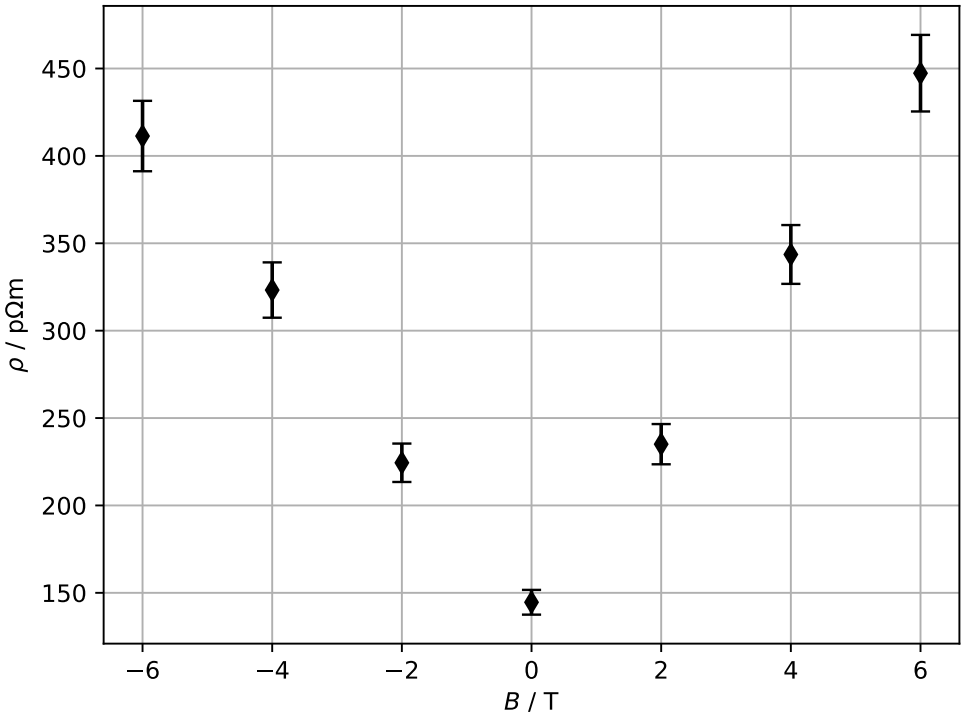


Figure 20: Electric resistivity for different magnetic fields at 30 K.

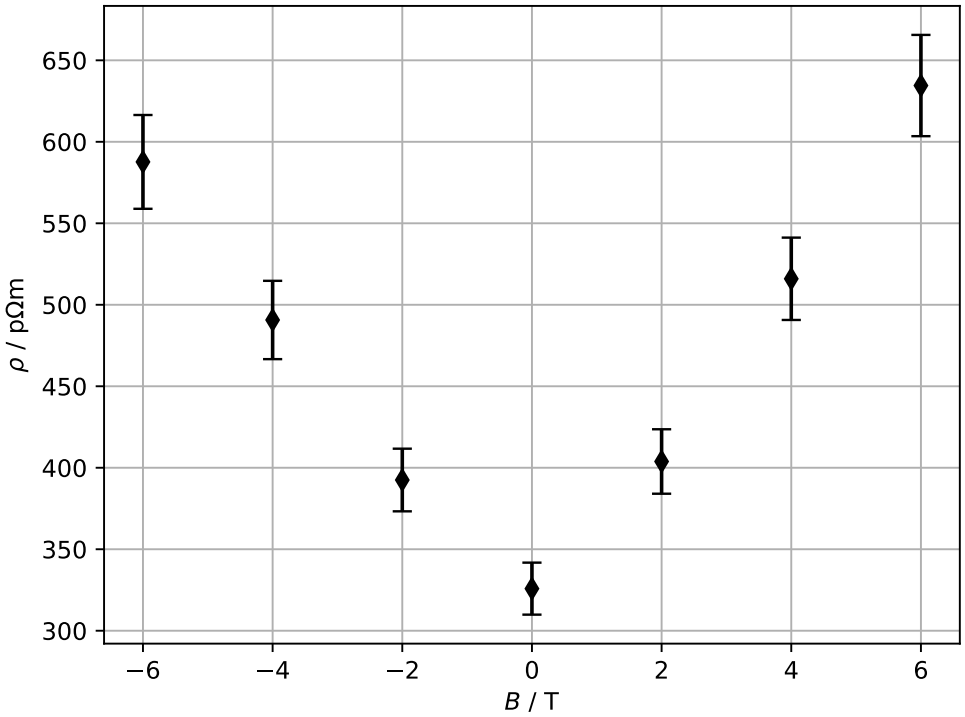


Figure 21: Electric resistivity for different magnetic fields at 40 K.

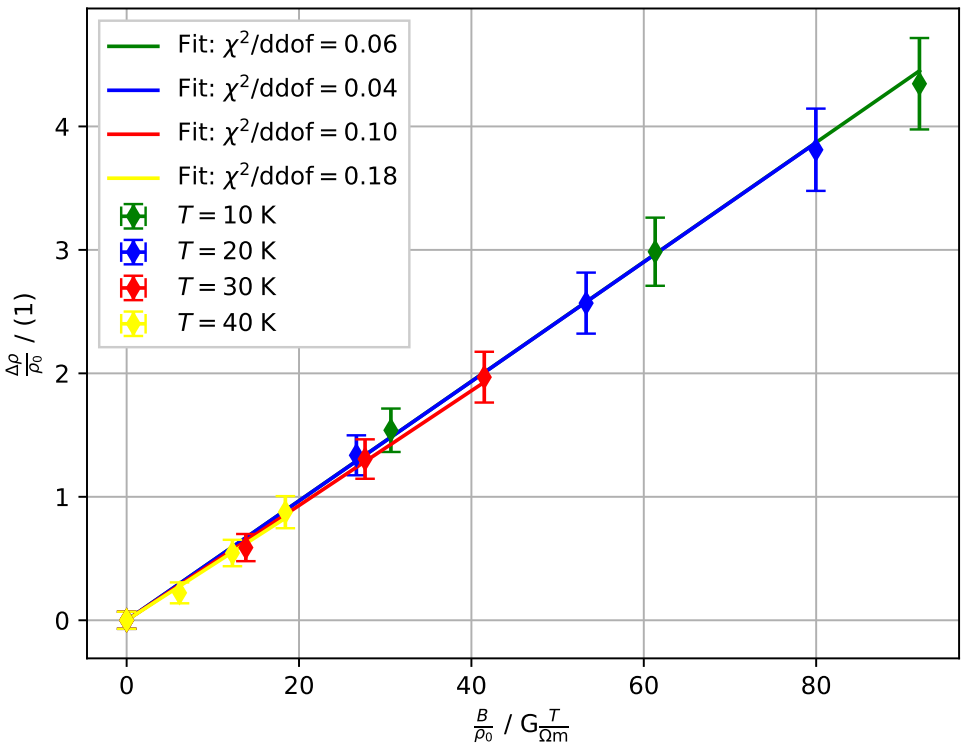


Figure 22: Kohler plot with $F(10\,\mathrm{K}) = 48.4(27)\,\frac{\mathrm{f\Omega m}}{\mathrm{T}}$, $F(10\,\mathrm{K}) = 48.4(28)\,\frac{\mathrm{f\Omega m}}{\mathrm{T}}$, $F(10\,\mathrm{K}) = 46.5(34)\,\frac{\mathrm{f\Omega m}}{\mathrm{T}}$ and $F(10\,\mathrm{K}) = 44.9(51)\,\frac{\mathrm{f\Omega m}}{\mathrm{T}}$.